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Urban wastewater contributes to the emergence of carbapenem-resistant *Klebsiella pneumoniae* (CRKP) in an urban receiving river in eastern India

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Abstract

The present study revealed the emergence of carbapenem-resistant *Klebsiella pneumoniae* (CRKP) and the associated driving factors in an urban river system surrounding Cuttack city, Odisha. The high contamination factor and contamination degree indicate poor water quality. The CRKP isolates showed 100% resistance against piperacillin, amoxicillin-clavulanic acid, piperacillin-tazobactam, ceftriaxone, ceftazidime, meropenem, and imipenem but less resistance to colistin (12.85%). Among the CRKP isolates, carbapenemase genes *bla*_{NDM}, *bla*_{OXA-48-like}, and *bla*_{KPC} were detected in 94.28%, 35%, and 10% of isolates, respectively. The resistance genes (*bla*_{NDM}, *bla*_{TEM}, and *bla*_{CTX-M}) were found to be significantly correlated with toxic metals (As, Cd, Co, Cu, Fe, Mn, Pb) ($P < 0.05$). Detection of virulence factors (yersiniabactin and aerobactin) and capsular serotypes (K1, K2, and K54 types) explain the pathogenicity of CRKP isolates. Enterobacterial repetitive intergenic consensus-PCR based molecular typing separated the CRKP strains into 13 clusters, of which VI and XI clusters showed similar resistance and virulence determinants, indicating the dissemination of clones from wastewater to the river system. Our results provide first-hand information on assessing risks to public health posed by the CRKP isolates and toxic metals in the Kathajodi River. Molecular surveillance of nearby hospitals for the prevalence of CRKP will help trace their transmission route.

Significance of the study: The biomedical and anthropogenic waste receiving urban rivers act as both the source and sink for drug-resistant bacteria including CRKP the top-ranked “WHO priority pathogen”. It is a critical public health concern for India, being the leader in CRKP infection. In this study, the detection of carbapenemase genes (*bla*_{NDM}, *bla*_{OXA-48-like}, *bla*_{KPC}), virulence factors, capsular serotypes and their correlation with metals, and dissemination of CRKP in river water explain their pathogenicity. Molecular surveillance of CRKP and their resistance pattern could be highly imperative for epidemiological study and controlling pathogens of the locality.

Keywords: antibiotics, toxic metals, carbapenem-resistant *Klebsiella pneumoniae* (CRKP), environment, carbapenemases, *bla*_{NDM}

Introduction

Antibiotic-resistant bacteria (ARB) carrying antimicrobial resistance genes (ARGs) are the major global public health concerns in the 21st century, owing to the continued rise in antibiotic usage in medical, domestic, agricultural, and veterinary sectors (WHO 2019). As a result, global antimicrobial use is anticipated to increase from 63 151 to 105 596 tons (67%) between 2010 and 2030 (Van Boeckel *et al.* 2014). Further, with a rising rate of antibiotic consumption in India in recent years (Klein *et al.* 2018), it has been predicted to be the next largest shareholder in total antimicrobial usage (Walia *et al.* 2019) in the absence of stricter rules for restricted antibiotics use in food products, animal husbandry, or the medical industry (GARP 2011, WHO 2014). Urban wastewater is a significant source of ARB, which get disseminated into the river or natural environment via effluents (Mukherjee *et al.* 2021). It poses a risk to public health due to the possibility of human exposure via bathing, drinking, and through the food chain. Carbapenem-resistant *Klebsiella pneumoniae* (CRKP) has been detected in the river from Tunisia (Chouchani *et al.* 2013), Switzerland (Zurfluh *et al.* 2013), Spain (Piedra-Carrasco *et al.* 2017), and Portuguese (Teixeira *et al.* 2020). Recently, a high prevalence of ARB and

ARGs has been reported in Indian rivers like Torsa, Gomati, Ganga, and Sabarmati, which is regarded as a severe public health issue (Mukherjee and Chakraborty 2006, Akhter *et al.* 2014, Chaturvedi *et al.* 2020, Ram and Kumar 2020). In addition, heavy metals are anthropogenic pollutants in river water that are known to promote bacterial resistance to antibiotics, perhaps paving their way to human hosts (Komijani *et al.* 2021). According to the Central Pollution Control Board (CPCB) national inventory of sewage treatment plants (STPs) issued in March 2021, urban India treats only 37% of the 72 368 megaliters per day (MLD) sewage, with around two-thirds of the wastewater contaminating the environment (<https://timesofindia.indiatimes.com/blogs/voices/wastewater-and-faecal-sludge-reuse-to-address-indias-water-and-food-security/>). Several studies have established that treated and untreated municipal wastewater, hospital effluents, and wastewater treatment plant effluents contribute significantly to the spread of antibiotic resistance in the aquatic environment (Sivalingam *et al.* 2019).

CRKP was designated as a critical priority pathogen by World Health Organization (WHO) in 2017 due to its clinical significance and remarkable ability to acquire resistance determinants (WHO 2017). According to the Center for Disease Dynamics, Economics & Policy (CDDEP) report, India

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has the highest rate of CRKP in the world (CDDEP 2015). Since CRKP can survive for weeks or even months in moist and dry environments, it is widely distributed and spread in natural habitats. These findings demand urgent surveillance due to increased pathogenicity in the river, which behaves as both source and sink for the emergence of drug-resistant bacteria through horizontal gene transfer.

Cuttack, Odisha, the Millennium City and the clinical hub of eastern India with a population of 610 189, is located at the bank of the Kathajodi River. Biomedical wastes are generated from 7 major specialized hospitals, 229 registered and over 300 unregistered nursing homes, clinics, and diagnostic centers and are collected in small, medium, and tertiary drains. This city generates an average of 65.44 MLD of wastewater per day, of which 58 MLD is discharged untreated or treated into the Kathajodi River. In addition, two STP units were installed for treatment. Despite the presence of STPs, the degree of pollution in the Kathajodi River is level C, indicating that its water cannot be used for any purpose, even in agricultural irrigation (<https://www.rrcodisha.org/wp-content/uploads/2020/08/Action-plan-for-Kathajodi-river-JULY-20.pdf>).

Kathajodi is among India's most polluted rivers (Mallick 2012) and is deemed unfit for human use (Express News Service 2019). Thus, for the first time, attempts are being made to find the prevalence of CRKP having the carbapenemase genes (*bla_{NDM}*, *bla_{KPC}*, *bla_{OXA-48-like}*, and *bla_{VIM}*) and the co-harboring ESBL genes (*bla_{SHV}*, *bla_{CTX-M}*, and *bla_{TEM}*) in the river system around Cuttack. Additionally, physicochemical characteristics were studied to determine the water quality, which may aid in addressing the immediate health risk posed by the community's reliance on the Kathajodi River. Furthermore, the association of the physicochemical parameters and toxic metals with the resistance determinants, and the possible drivers of resistance in the river system are also established. We hypothesize that the anthropogenic activity surrounding Cuttack city could significantly affect the dissemination of CRKP in the river system and help in the migration of the resistance genes downstream.

Results and discussion

Physicochemical parameters

The surface water samples collected from different sites (K1, K2, K3, K4, K5) of Cuttack city were assessed for various physicochemical parameters (Table S1). The pH ranged from 7.00 ± 0.01 to 7.7 ± 0.01 , implying the water to be neutral to slightly alkaline and within the permissible limit of the Bureau of Indian Standards (BIS), WHO, and United States Environmental Protection Agency (US EPA). Similarly, the temperature of all sites did not show any high temporal variation. Total dissolved solids (TDS) concentration at five sites varied from 120.38 ± 0.12 to 389.35 ± 0.23 mg l⁻¹, with the K2 site having the highest TDS of 389.35 ± 0.23 mg l⁻¹ and the K1 site having the lowest of 120.38 ± 0.12 mg l⁻¹. Similarly, the highest electrical conductivity (EC) (599.00 ± 7.22 μ mho cm⁻¹) was recorded from the K2 site and the lowest (185.2 μ mho cm⁻¹) from the K1 site. Dissolved oxygen (DO) was highest in the K1 (185.2 ± 4.92 mg l⁻¹) location, whereas the lowest DO was recorded at K3 (0.02 ± 0.001 mg l⁻¹). The biological oxygen demand (BOD) and chemical oxygen demand (COD) exceeded the limits established by international standard-setting organizations (Table S1). The physicochemical analysis of surface water plays an important

role and directly impacts water quality. Moreover, aquatic life is also affected by water quality degradation (Ali *et al.* 2016).

Metal concentration in the water sample

Metals in the water samples are shown in Table S2. Contents of several metals, including sodium (Na), potassium (K), copper (Cu), chromium (Cr), zinc (Zn), arsenic (As), mercury (Hg), barium (Ba), nickel (Ni), selenium (Se), silver (Ag), molybdenum (Mo), titanium (Ti), beryllium (Be), vanadium (V), and cobalt (Co), were within the prescribed limit for all the sampling sites as per BIS, WHO, and US EPA. In contrast, concentrations of iron (Fe), manganese (Mn), and aluminum (Al) in all locations exceeded the recommended guidelines except for the K1 site. The highest Fe, Mn, and Al concentration was recorded at the K2 site as 2243.07 ± 173.9 , 663.56 ± 36.06 , and 1604.58 ± 82.96 μ g ml⁻¹, respectively. The concentration of cadmium (Cd) at K3 and K4 was higher than the permissible limit, while lead (Pb) concentration was above the prescribed limit of BIS and WHO standard at the K2 site (11.13 ± 2.64 μ g ml⁻¹), as shown in Table S2. The higher concentration of Fe, Mn, Cd, Pb, and Al, in wastewater and river water, is commonly found in other Indian rivers Tami-raparan and Ganga (Kar *et al.* 2008, Magesh *et al.* 2017). The Fe concentration in water samples was found to be exceptionally higher, possibly due to the direct discharge of various effluents from textile, milling, plating, and surface finishing industries surrounding the city. It promotes the absorption of other toxic metals such as Cu and Pb (Tchounwou *et al.* 2012) involved in hemochromatosis (Hussain *et al.* 2017), whereas Pb affects the central nervous system, gastrointestinal tract, teratogenic effect, and kidney dysfunction (Ferner 2001, Udedi 2003). Furthermore, Al causes dementia and induces certain cancers (Mandeng *et al.* 2019). High concentrations of toxic metals at K3 and K4 sites could be due to increased anthropogenic activity, agricultural runoff, sanitary landfills, and domestic wastewater effluents. These metals are known to be hazardous and could affect water supplies. In contrast, the upstream region exhibited decent quality in trace element concentration.

Contamination factor, contamination degree, and pollution load index

Compressive measurement of the toxic metal pollution in the surface water is determined from contamination factor (CF), contamination degree (CD), and pollution load index (PLI) (Table 1). The CF for Cd was most extreme at K3, followed by K4, K5, and K2 sampling sites. Similarly, CF for Mn was highest at the K2 site, which was 6.847, followed by K3, K4, and K5 sites (Table 1). Metals such as Ag, As, Cr, Co, Hg, Ni, and Zn belonged to the CF class 1 at all sampling sites, whereas Fe, Cu, and Pb belonged to class 2 at K2, K3, and K4 sites. The CF value for all the metals at the K1 site was under class 1. The overall results indicated that river water was highly polluted with heavy metals such as Cd, Cu, Mn, Pb, and Fe due to wastewater. A similar analysis was also reported in the Karnaphuli River in Bangladesh, which has the highest CF by Cd heavy metal (Ali *et al.* 2016). CD suggested that the wastewater (K2 and K3) falls in class 3 and class 4, contaminating the river water (K4), which falls in class 4 (Table 1). The PLI ranged from 0.065 to 0.811, showing that toxic metal contamination was present in the surface water (Table 1). The lowest CD and PLI values were recorded at

Table 1. Metal CF, CD, and PLI of river water and wastewater.

Toxic metals	Sampling sites				
	K1 (CF)	K2 (CF)	K3 (CF)	K4 (CF)	K5 (CF)
Ag	0.69	0.081	0.104	0.277	0.172
As	0.137	0.864	0.639	0.524	0.58
Cd	0.008	2.218*	22.31***	21.113***	2.285*
Cr	0.311	0.401	0.369	0.229	0.172
Co	0.067	0.46	0.327	0.407	0.511
Cu	0.016	3.73**	2.243*	1.566*	0.967
Fe	0.122	2.635*	1.191*	1.386*	1.078*
Hg	0.013	0.024	0.027	0.016	0.07
Mn	0.083	6.847***	6.236***	4.144**	3.238**
Ni	0.239	0.672	0.484	0.252	0.198
Pb	0.008	2.812*	1.242*	1.15*	0.507
Zn	0.938	0.904	0.452	0.307	0.138
Contamination degree (CD)	1.78	21.653	35.629	31.375	9.921
Pollution Load index (PLI)	0.065	0.811	0.728	0.627	0.448

Notes: No asterisk, (low CF; class 1); *, (moderate CF; class 2); **, (considerable CF; class 3); ***, (high CF; class 4).

the K1 site because of low anthropogenic activity. However, increased anthropogenic activity was due to population load and direct discharge of wastewater (K2 and K3) into the river catchment (K4) and downstream (K5), resulting in the highest CD and PLI values. A significant correlation between the metals may indicate their release from the same source, mutual dependency, and identical behavior during their transmission in the river system (Ali *et al.* 2016). High CD and PLI values indicate poor water quality and provide critical information to decision-makers regarding the pollution status.

Enumeration of total bacteria and CRKP

The total coliform (TC) contamination was higher at sites K2, K3, and K4, i.e. $160.00 \pm 1 \times 10^3$ MPN 100 ml⁻¹ and lower at sites K1 and K5, i.e. $7.83 \pm 0.94 \times 10^3$ and $11.78 \pm 2.86 \times 10^3$ MPN 100 ml⁻¹, respectively. Similarly, the fecal coliform (FC) count was highest at sites K2, K3, and K4, i.e. $160.00 \pm 1 \times 10^3$ MPN 100 ml⁻¹ and lowest at site K1 ($2.22 \pm 0.45 \times 10^2$ MPN 100 ml⁻¹). The total number of bacteria found in samples from river water and wastewater is represented in Table S3. The highest number of *Klebsiella* spp. resistant to carbapenem was noted at K2 ($7.21 \pm 0.38 \times 10^8$ CFU ml⁻¹), followed by K3 ($5.1 \pm 0.52 \times 10^8$ CFU ml⁻¹), K4 ($1.99 \pm 0.51 \times 10^8$ CFU ml⁻¹), and K5 ($8.1 \pm 0.74 \times 10^7$ CFU ml⁻¹) sites. No carbapenem-resistant *Klebsiella* sp. was found at the K1 sampling site (Table S3). A total of 243 *Klebsiella* spp. (K2, $n = 109$; K3, $n = 67$; K4, $n = 54$, and K5, $n = 13$) were picked from HiCrome™ *Klebsiella* Selective Agar (HiMedia) plate supplemented with meropenem. Out of these 168 isolates (69.13%, $n = 243$) showed stable carbapenem resistance. Among these, 138 isolates were identified as *K. pneumoniae* using VITEK 2 system and further confirmed by *Khe* gene PCR amplification. After enterobacterial repetitive intergenic consensus (ERIC)-PCR analysis, a total of 70 unique CRKP isolates were selected for further studies. The highest unique CRKP clones were obtained at K3 site ($n = 28$) followed by K4 ($n = 22$) and K2 ($n = 13$) sites. The least unique clones were found at downstream location K5 ($n = 7$). CRKP in water bodies was reported by Ekwanzala *et al.* (2019), which corroborates the finding of this study.

Antibiotic resistance profiles and MAR index

A total of 47 (63.15%) and 23 (32.85%) CRKP isolates showed multidrug resistance (MDR) and extremely drug resistance (XDR) characteristics, respectively. The highest percentage of XDR isolates was found at K3 and K4 sites. Detection of MDR and XDR determinants explains the extent of deterioration of water quality in Kathajodi River when compared with other rivers of India (75.62%) (Chaturvedi *et al.* 2021), Nigeria (65.8%) (Obayiuwana *et al.* 2018), South China (87.5%) (Su *et al.* 2012), and Ethiopia (75.8%) (Berhane and Meniyamer 2015). All the CRKP isolates from each sampling site showed 100% resistance to β -lactam (PI), β -lactam/ β -lactamase inhibitor combinations (AMC; PIT), third-generation cephalosporin (CTR; CAZ), and carbapenem (MRP; IMP). In addition, these are the most frequently prescribed antibiotics to treat nosocomial infections (Korzeniewska *et al.* 2013, Kotlarska *et al.* 2015, Pazda *et al.* 2019, Chaturvedi *et al.* 2021). About 80% and 92% of CRKP isolates of K2 and K5 sites showed resistance to fluoroquinolone (CIP; LE) and aminoglycoside (AK, GEN), respectively. However, some isolates at K2 (15.38%), K3 (17.85%), and K4 (9.09%) sites showed resistance to colistin, and all the isolates were susceptible to glycylcycline (TGC) (Fig. 1a). In this study, the isolates showed resistance to several groups of antibiotics, which corroborated with that of other Indian river system (Chaturvedi *et al.* 2021). The sampling sites (K2, K3, K4, and K5) had a multiple antibiotic resistance (MAR) index greater than 0.2, indicating a higher rate of antibiotic contamination and potential areas for emerging resistant bacteria to contaminate Cuttack city. Our results also concatenated the Ganga River system, where the MAR index was >0.2 (Chaturvedi *et al.* 2020). The expansion of the resistant domain can be due to the rapidity with which unregulated anthropogenic activities have progressed in conjunction with the excessive use of antibiotics. Patro *et al.* (2018) reported carbapenem-resistant *Enterobacteriaceae* from ventilator-associated pneumonia at a tertiary care hospital in Cuttack city (Patro *et al.* 2018). This suggests that the extensive use of β -lactam antibiotics in tertiary care hospital might have contributed to the presence of a large number of β -lactam-resistant bacteria in water bodies resulting in their accumulation via medical effluents.

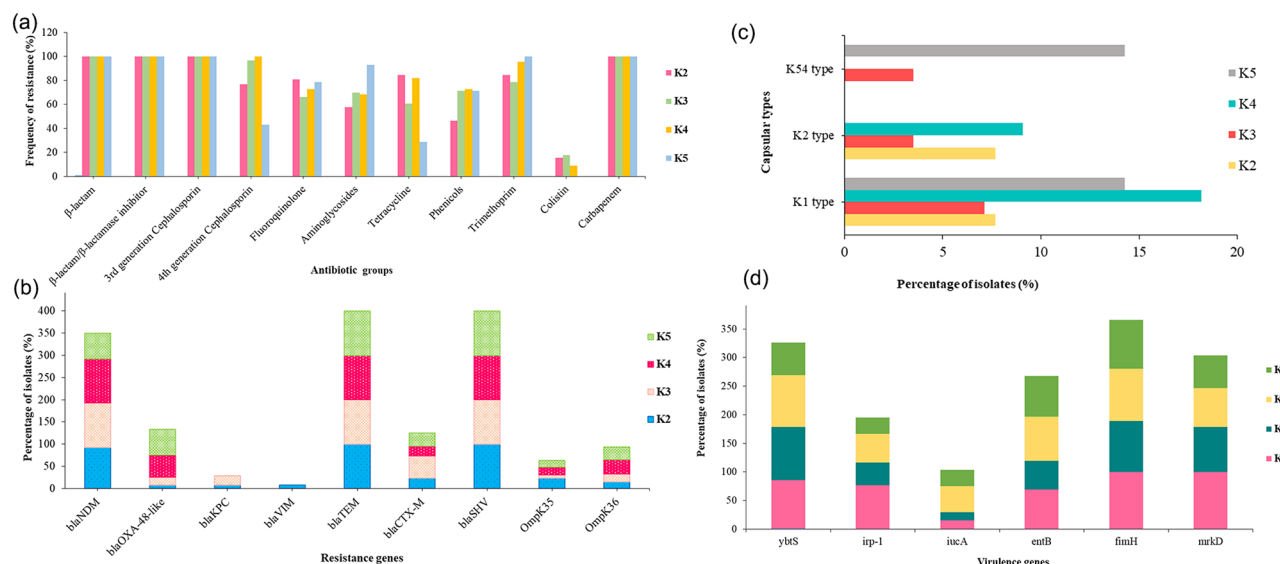


Figure 1. (a) Distribution of antibiotic resistance pattern. (b) Prevalence of carbapenem, ESBL, and outer-membrane porin genes. (c) Prevalence of capsular serotypes and (d) virulence factors of CRKP isolates in river system, Cuttack city, Odisha.

Characterization of carbapenemase, ESBL, and outer-membrane porin genes

To further identify the resistance mechanism involved in these bacteria, we analyzed a panel of genes that controls the manifestation of antibiotic resistance. Among 70 CRKP isolates, 100% prevalence of *bla*_{NDM} was observed in K3 and K4 sites, while in K2 and K5, it was 92.3% (12/13) and 57.14% (4/7), respectively. Similarly, the *bla*_{OXA-48-like} gene was identified at K4 (11/22; 50%) sampling site, followed by K5 (4/7; 57.14%), K3 (5/28; 17.85%), and K2 (1/13; 7.69%) sites. The *bla*_{KPC} gene was identified in six isolates at K3 (6/28; 21.42%) and one isolate at K2 (1/13; 7.69%). The *bla*_{VIM} was present in only one isolate at the K2 site. Additionally, most CRKP isolates co-harbored ESBL genes (*bla*_{TEM}, *bla*_{CTX-M}, and *bla*_{SHV}). Out of these genes, *bla*_{TEM} and *bla*_{SHV} were found in all CRKP isolates. The *bla*_{CTX-M} was mostly identified at K3 (14/28; 50%), followed by K5 (2/7; 28.57%), K2 (3/13; 23.07%), and K4 (5/22; 22.72%) sites. The distribution of carbapenemase and ESBL genotypes among CRKP isolates at various sites is depicted in Fig. 1b. The co-occurrence of carbapenemase and ESBL resistance genes detected from different environment types (hospital sewage, effluents, and receiving rivers) across the globe supports the present finding (Singh et al. 2019b, Smyth et al. 2020, Zhang et al. 2020). The high quantity of β -lactam antibiotics in water bodies accumulated through medical effluent might have caused selective pressure on harboring bacteria to produce β -lactamase through the overexpression of respective genes (Singh et al. 2016). Outer-membrane porins (*ompK35* and *ompK36*) are protein channels in *K. pneumoniae* that facilitate small hydrophilic molecules such as iron, nutrients, and antibiotics (Tsai et al. 2011, Ferreira et al. 2019). In this study, *ompK36* was most prevalent at all sites (K2, K3, K4, and K5), whereas *ompK35* was least found at K2 (3/13; 23.07%), K4 (4/22; 18.18%), K5 (1/7; 14.28%), and K3 (2/28; 7.14%) sites. More than 60% of the isolates lacked *ompK* genes that contribute resistance to antibiotics such as ceftazidime, carbapenem, ciprofloxacin, and chloramphenicol (Indrajith et al. 2021). However, 40% of the isolates have *ompK35* and *ompK36*, responsible

for antibiotic resistance due to point mutations or disruption of the porin protein-coding sequence (Hamzaoui et al. 2018).

Detection of capsular serotypes and virulence factors

Capsular serotype was mostly associated with virulence and pathogenicity of CRKP. From 70 CRKP isolates, 14 (20%) were typeable serotypes, and 56 (80%) did not belong to capsule types determined in this study. Three different capsular serotypes K1, K2, and K54 were found in this study. The K1 type (4/22; 18.18%) and K2 type (2/22; 9.09%) were majorly isolated from the K4 site. Similarly, K1 type (1/7; 14.28%) and K54 type (1/7; 14.28%) were mostly identified in the isolates from the K5 site. The CRKP isolate of K3 site harbored K1 type (2/28; 7.14%), K2 type (1/28; 3.57%), and K54 type (1/28; 3.57%) serotypes. The other serotypes K5, K57, K20 were not detected. The prevalence of capsular serotypes at various sites is depicted in Fig. 1c. The capsule has been understood to aid in the invasion and colonization of human hosts, adding to the pathogenicity (Ssekatawa et al. 2021). In this study, the CRKP isolates were identified as either K1, K2, or K54 serotypes, known to be virulence determinants of liver abscesses and pulmonary failure-related infections in human beings (Barati et al. 2016, Hsu et al. 2016, Shankar et al. 2018, Zhang et al. 2018, Ferreira et al. 2019, Liu et al. 2019, Sahoo et al. 2019, Hasani et al. 2020).

This study also screened the virulence factors such as siderophore, host adherence, and iron uptake ability. The siderophore virulence factor *ybtS* was found to be highest at the K3 site (26/28, 92.85%), followed by K4 (20/22, 90.9%), K2 (11/13, 84.61%), and K5 (4/7, 57.14%) sites. The other siderophore genes *iucA* and *entB* were most prevalent in isolates from K4 (10/22, 45.54%; 17/22, 77.27%) samples followed by K5 (2/7, 28.57%; 5/7, 71.42%), K2 (2/13, 15.38%; 9/13, 69.23%), and K3 (4/28, 14.28%; 14/28, 50%) sites (Fig. 1d). The iron uptake virulence genes *iutA* and *iroB* were not found in any of the CRKP isolates that are detected in hypervirulent *K. pneumoniae* (Russo et al. 2015).

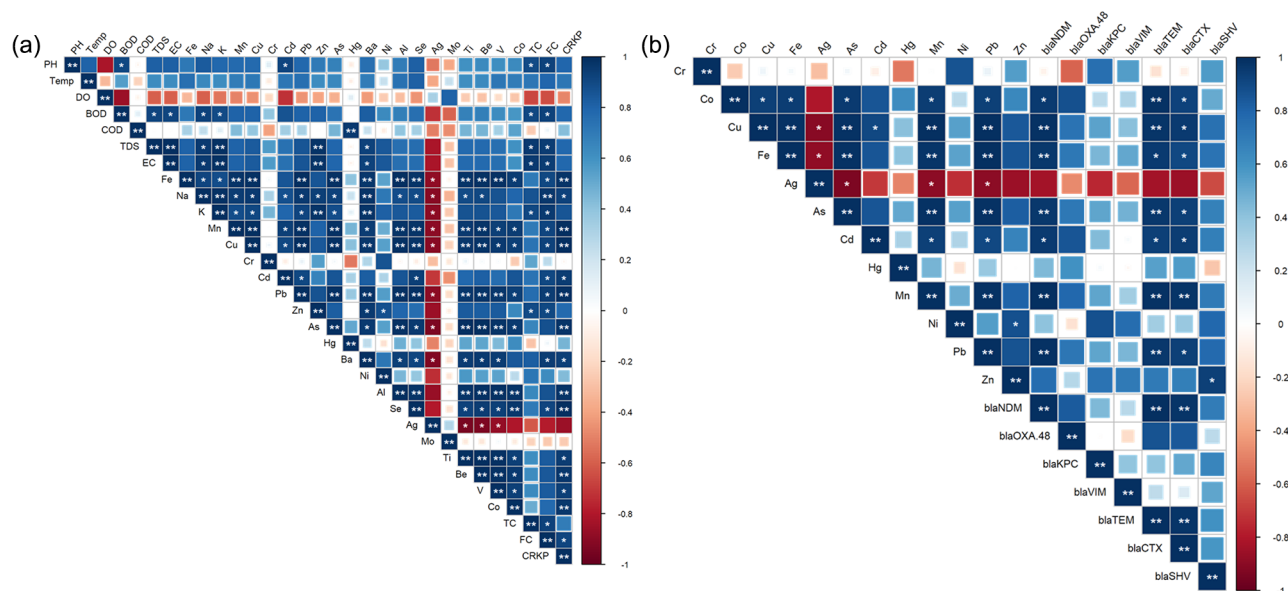


Figure 2. Pearson correlation plot (a) between concentration of physicochemical parameters, metals, and CRKP (CFU ml⁻¹) and (b) between concentration of toxic metals and resistance genes. The dark blue colors indicate high positive correlation and dark red colors indicate high negative correlation. * $P < .05$, ** $P < .01$.

Most of the hypervirulent strains were obtained from catchment locations and downstream of the river. The host adherence virulence genes *fimH* and *mrkD* were found to be highest at K2 (13/13, 100%) followed by K4 (20/22, 90.9%; 15/22, 68.18%), K3 (25/28, 89.28%; 78.57%), and K5 (6/7, 85.71%; 4/7, 57.14%) sites, whereas *KpcA* was not detected. The capsular upregulation genes (*rmpA* and *rmpA2*) were not amplified in any isolates, as also observed by Rolbeikie *et al.* (2021). They found that *ybtS*, *mrkD*, *entB*, and *iutA* were most abundant in carbapenem-resistant *Klebsiella* spp. isolated from wastewater and river water of Poland. These virulence factors are identified in nearly all isolates, and are involved in the pathogenesis of the bacterial infection (Hennequin and Robin 2016). The hypervirulent factor aerobactin (*iucA*) was identified in 25.71% of the *K. pneumoniae* isolates but *iutA* genes were not detected. The *iutA* was responsible for the transport of aerobactin into the periplasm. However, with the absence of this gene, yersiniabactin and enterobactin may have compensated for the deletion of *iutA* (Ling *et al.* 2013).

Correlation analysis between physicochemical parameters, metals, and CRKP (CFU ml⁻¹)

The abundance of CRKP isolates positively correlated with toxic metals (Fe, Na, Mn, Cu, Cd, Pb, As, Ba, Al, Se, Ti, Be, V, and Co). This finding reflects that these metals serve as growth factors for developing antibiotic resistance against CRKP-like opportunistic pathogens, which is similar to the report described by Nguyen *et al.* (2019). However, Na, Cu, As, Se, Ti, Be, V, and Co were lower than the permissible limit but strongly correlated with CRKP isolates. It is possibly due to metals being absorbed by the cell wall for bacterial cellular processes (Deng *et al.* 2020). Similarly, FC showed a positive correlation with Fe, Na, K, Mn, Cu, Cd, Pb, Zn, Ba, and Se, whereas TC showed a positive correlation with only K and Zn. Both FC and TC showed a positive correlation with pH, BOD, TDS, and EC ($P < .05$). A correlation analysis revealed

a positive relationship between physicochemical parameters (TDS and EC) and metals (Na, K, Zn, and Ba) (Fig. 2a; Table S4). The CRKP isolates were not showing any significant correlation with physicochemical parameters. This finding reflects that these metals serve as growth factors for developing antibiotic resistance against CRKP-like opportunistic pathogens.

Correlation between ARGs and toxic metals

The correlations among the metals and abundance of carbapenemases (*bla*_{NDM}, *bla*_{OXA-48-like}, *bla*_{KPC}, *bla*_{VIM}) and ESBL (*bla*_{TEM}, *bla*_{CTX-M}, *bla*_{SHV}) were evaluated by the Pearson's correlation coefficients. There was a high positive correlation between *bla*_{NDM} and *bla*_{TEM} ($r = .993$, $P < .01$), *bla*_{NDM} and *bla*_{CTX-M} ($r = .975$, $P < .01$), and *bla*_{CTX-M} and *bla*_{TEM} ($r = .978$, $P < .01$). The result showed that three genes (*bla*_{NDM}, *bla*_{TEM}, *bla*_{CTX-M}) displayed a significant correlation with toxic metals As, Cd, Co, Cu, Fe, Mn, and Pb ($P < .05$, .01), which suggests that the heavy metals could be a distinguishable factor responsible for influencing the residing β -lactam genes and helping in distribution and propagation within CRKP strains. *bla*_{SHV-like} gene was positively correlated with Zn metal ($r = .933$, $P < .05$) (Fig. 2b; Table S5). It is worth noting that the antibiotics and heavy metal resistant genes were identified in the same position of chromosomes or mobile genetic elements because bacteria might easily acquire multiple resistance properties (Xu *et al.* 2017). Reports on the facilitation of metals in transmitting and proliferation of antibiotic resistance genes through conjugative plasmid transfer further support our findings (Wang *et al.* 2021).

ERIC-PCR analysis

ERIC-PCR analysis revealed a genetic linkage varying from 50% to 100% between CRKP isolates. The genetic diversity of 70 CRKP isolates exhibited 56 distinct ERIC profiles grouped into 13 clusters (C-I to C-XIII). The minimum similarity between clusters I and IV was 40% and maximum similarity

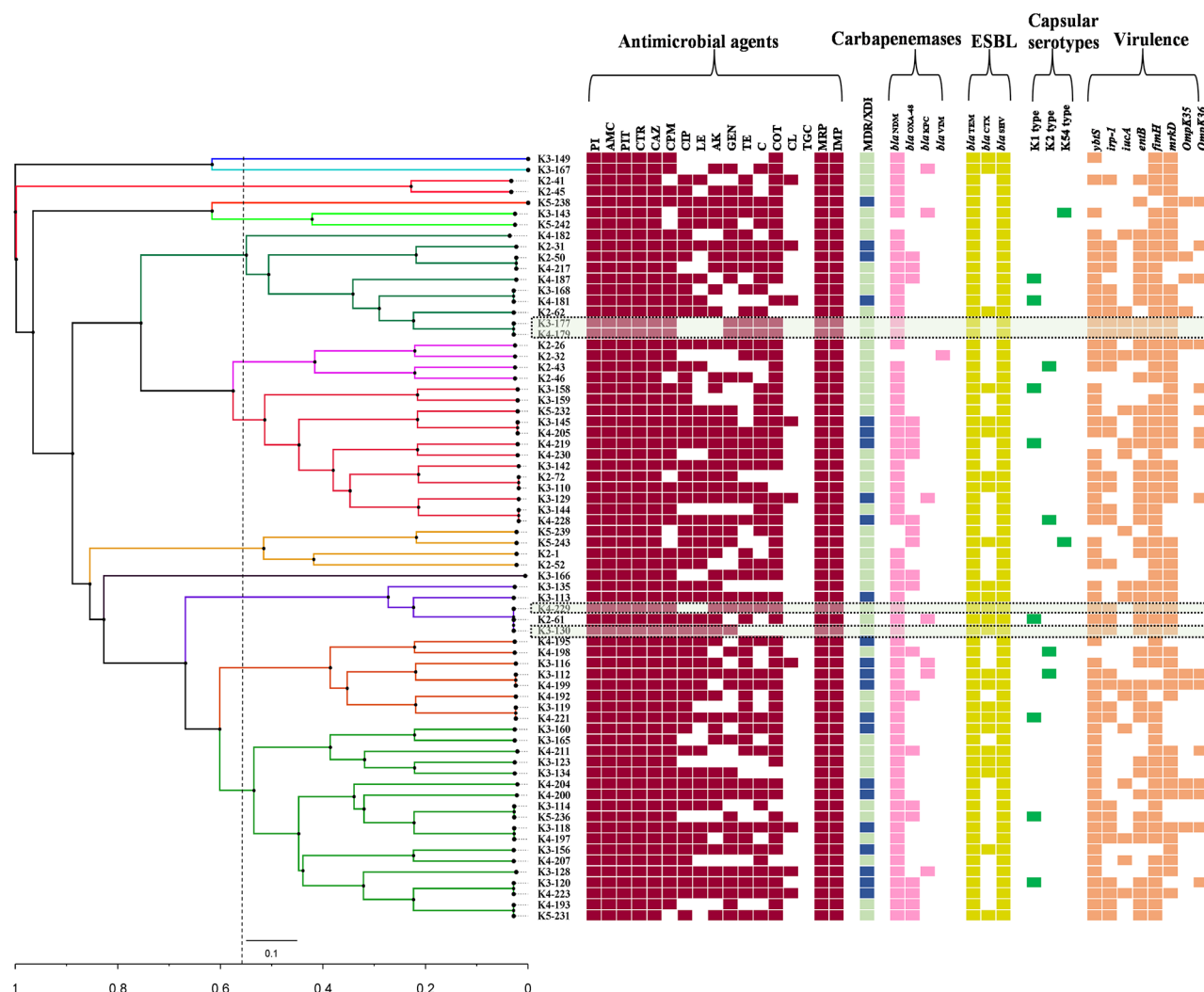


Figure 3. Phylogenetic tree showing the relationship between CRKP isolates. Colors in filled boxes indicate as follows: red, antibiotic resistance; light green, MDR; blue, XDR; pink, carbapenemases; yellow, ESBL genes; green, capsular serotypes; and orange, virulence factors. The dotted rectangular boxes indicate CRKP isolates having similar resistance genes and virulence factors.

between clusters I and VI was 78%. The average similarity value between the clusters was 68%. The largest cluster (C-XIII) had 18 CRKP isolates from K3 (50%), K4 (38.88%), and K5 (11.11%). The second-largest cluster (C-VIII) consisted of 13 CRKP isolates from K3 (53.84%), K4 (30.76%), K5 (7.69%), and K2 (7.69%) sites. A higher level of XDR isolates was found in C-XIII (44.44%) than in C-VIII (38.46%). The significant difference between the clusters (C-VIII and C-XIII) was the occurrence of carbapenemase-producing genes. Almost 38.88% isolates from C-XIII and 38.46% from C-VIII harbored *bla*_{OXA-48-like}. Similarly, *bla*_{KPC} was harbored in 7.69% and 5.56% isolates from C-VIII and C-XIII, respectively. Cluster VI consisted of 10 CRKP isolates from K4 (50%), K2 (30%), and K3 (20%) sites. Three CRKP isolates from the K3 site and one from K5 site were not phylogenetically connected to the other cluster and thus classified as a distinct cluster (Nos. I, II, IV, and X). The remaining clusters (Nos. V, VIII, XI, and XII) were composed of phylogenetically linked CRKP isolates from wastewater (K2 and K3) and river water (K4 and K5). Some isolates in clusters VI, VIII, XI, XII, and XIII showed similar genetic patterns but originated from

different sampling sites (Fig. 3). All the isolates were segregated into 13 clusters, of which clusters VI and XI (K3, 130 and K4, 229; K3, 177 and K4, 179) showed similar resistance and virulence determinants. This result implies dissemination from wastewater (K3) to river water (K4), which corroborates with Zhang *et al.* (2018) and Ali and Al-Kakei (2019). Variation in the ERIC pattern between the isolates is not highly associated with the presence of resistance or virulence genes, which is consistent with the clinical isolates reported by Wasfi *et al.* (2016) and Ferreira *et al.* (2019). Moreover, the sampling site receives bacterial populations other than CRKP, which have not been explored in this study. These sites serve as a perfect reactor for the exchange of genetic material intraspecifically as well as interspecifically. Furthermore, the plasticity of *K. pneumoniae* genomes has also been well documented (Ekwanzala *et al.* 2019). The consensus of all these factors might have a key role in the high variability of *K. pneumoniae* isolates identified in this study and needs further investigation. This molecular typing has been popularly employed to detect bacteria and gene transmission because of its cost-effective, rapid, and discriminatory ability over other molecular typ-

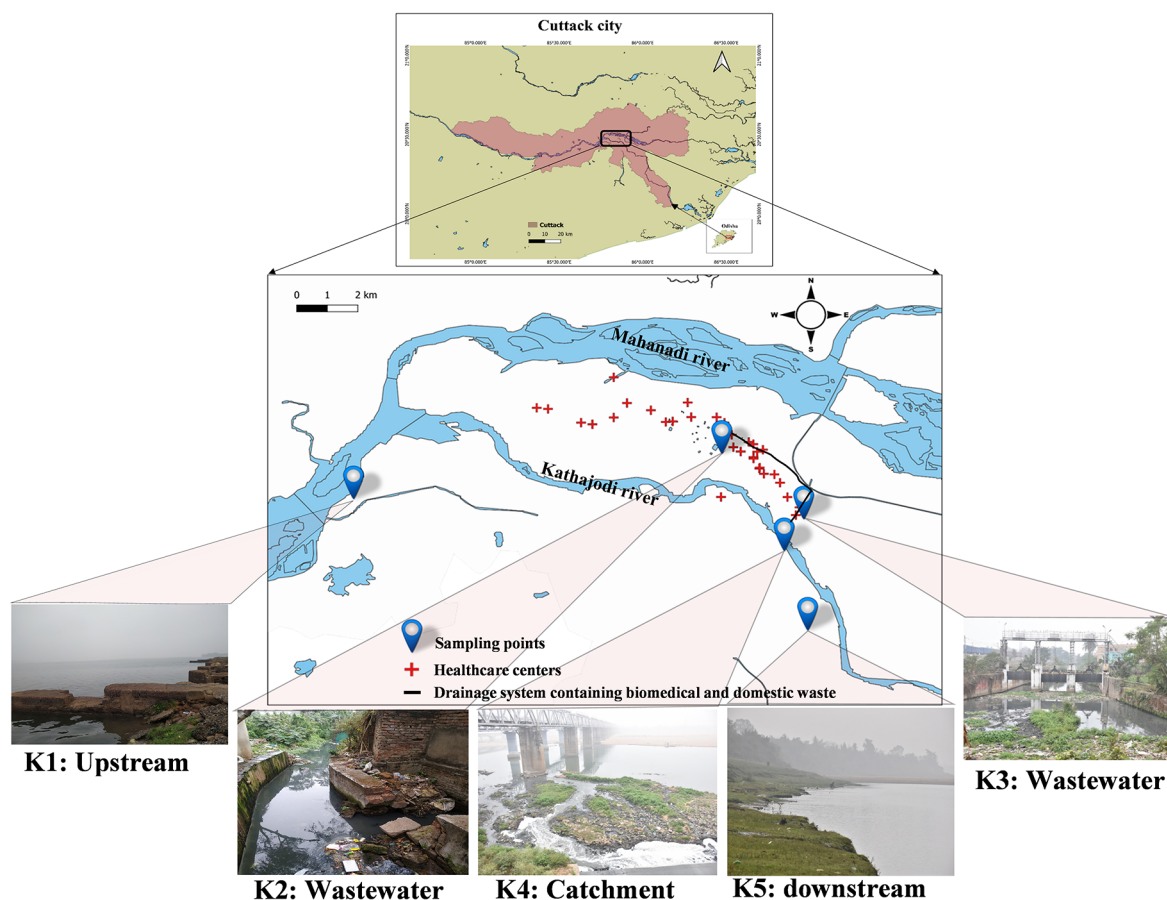


Figure 4. Map showing sampling sites in river system surrounding Cuttack city (pink color, top), Odisha (map source: QGIS software version 3.16.2, <https://qgis.org/en/site/forusers/download.html>). Location of sampling site (black rectangle); five blue color place marks represent sampling sites (K1–K5).

ing options such as Pulsed-field Gel Electrophoresis (PFGE) and Multilocus sequence typing (MLST) (Osińska *et al.* 2017, Zarei *et al.* 2018). However, routine molecular surveillance of the river around the city at regular intervals for a longer period covering a larger community through large-sized sampling will continue through the light into disseminating key resistance determinants. This could further help predict the outbreak of CRKP in the locality. In addition, quantifying antibiotic residues in the river water would help establish their correlation with the most prevalent resistant genes.

Conclusion

The findings emphasized that direct discharge of wastewater from healthcare units, domestic units, and industries into the river results in genetic material exchange via horizontal gene transfer and the spread of CRKP pathogens by providing an ideal environment. A significant correlation of metals with CRKP isolates could promote the growth of such pathogenic bacteria. Additionally, the discovery of threatening CRKP strains in the river highlights the critical importance of establishing initiatives to raise public awareness and frequent monitoring of surface water to forecast resistance levels, which could help direct antibiotic use and resistance management. Stringent strategies should be implemented by local functionaries for wastewater treatment to avoid resistant bacteria and genes entering the food chain.

Materials and methods

Sampling site and sample collection

The research was carried out in the river system surrounding Cuttack city, regarded as the silver city of India for excellent traditional silver filigree work by the local community. Cuttack is located at delta formed by the river Mahanadi and Kathajodi, covering a geographical area of 3932 km with a population of 610 189. Surface water samples were collected from five different sites on 17 March 2020 from upstream (Mahanadi River) and ~15 km downstream of the Kathajodi River (Fig. 4). River water samples were taken from upstream location from the Cuttack city (K1, 20° 27' 40.7" N; 85° 45' 21.1" E), river catchment area where wastewater is directly discharged into the river (K4, 20° 26' 39.1" N; 85° 53' 27.4" E), and downstream location (K5, 20° 24' 46.1" N; 85° 54' 24.2" E). In addition, two untreated wastewater samples (K2, 20° 28' 34.7" N; 85° 52' 16.9" E and K3, 20° 27' 16.8" N; 85° 53' 49.4" E) were taken from the city's main drainage system, which is directly connected to the river catchment. Except for K1 sampling site, the aquatic plants are not observed on the surface of the water at K2, K3, K4, and K5 sampling sites. The sample was collected in triplicates at each location while maintaining a distance of 100 m from each other. These samples were combined to make a single representative sample of each site. About 500 ml of water samples were collected from the top 0.5 m of the river surface in a ster-

ile amber-colored bottles and transported to the laboratory in the ice box. All the water samples were stored at 4°C before being processed. On the day of collection, the samples were processed for physicochemical, metal, and bacteriological analysis.

Physicochemical analysis

General water quality parameters such as pH and temperature were measured using a pH meter (Adwa AD12) and a digital thermometer (MT-222 DigiFlexi, Dr. Morepen, India). Other parameters, including BOD, COD, DO, TDS, and EC, were measured as described by American Public Health Association (APHA 2012) at the State Water Testing Laboratory, Government of Odisha, and results were interpreted with Indian Standard for Drinking Water as per BIS (Bureau of Indian Standards [BIS] 2012), WHO (Fawel *et al.* 1996), and the US EPA (EPA 2004). Furthermore, TC and FC were calculated using the multiple tube fermentation process (APHA 2012) (Method no. 9221, APHA 2012).

Metal analysis

About 21 metals, including Fe, Na, K, Mn, Cu, Cr, Cd, Pb, Zn, As, Hg, Ba, Ni, Al, Se, Ag, Mo, Ti, Be, V, and Co, were analyzed by inductively coupled plasma mass spectroscopy (NexION 2000, Perkin Elmer) and interpreted with the standard limits as recommended by EPA, WHO, and BIS to draw reliable conclusions.

Estimation of toxic metal contamination

CF, CD, and PLI are the indexes used in this study to assess surface water contamination with toxic metals.

Contamination factor

CF was used to monitor toxic metal contamination in water samples over time (Hussain *et al.* 2020). It is the ratio of the average concentration of each metal in the sample to the background concentration of the same metal and is described in the following equation:

$$CF_i = C_i/C_b$$

C_i and C_b are the measured concentration of the metal and the background concentration of the specific metal. Such comparisons can be made at the local level. The average concentration of the river Mahanadi (Bureau of Indian Standards [BIS] 2012) was used as the representative local background concentration. The CF can be classified according to their values ranging from 1 to 6, as follows: if $CF < 1$, class 1 or low pollution; $1 < CF < 3$, class 2 or moderate pollution; $3 < CF < 6$, class 3 or considerable pollution; $CF > 6$, class 4 or very high pollution (Ali *et al.* 2016).

Contamination degree

CD was used to provide the degree of overall metal contamination in surface water at particular sites and determined by the sum of all CF values at a particular sampling site. It was classified into four categories, i.e. class 1 (low CD: $CD < 6$), class 2 (moderate CD: $6 \leq CD < 12$), class 3 (considerable CD: $12 \leq CD < 24$), and class 4 (high CD: $CD \geq 24$) (Rahman *et al.* 2014).

Pollution load index

PLI was used to assess the toxicity of a site under the influence of metal pollution (Thomilson *et al.* 1980). It was defined as the n th root of the product of CF of all the metals studied:

$$PLI = (CF_1 \times CF_2 \times CF_3 \times \dots \times CF_n)^{1/n}$$

where n is the number of metals, $PLI = 0$ is perfection, $PLI = 1$ is baseline level, and $PLI > 1$ is progressive deterioration (Malsiu *et al.* 2020).

Enumeration of total bacteria and CRKP

A microbiological study was performed in the laboratory within 24 h of sample collection. All water samples were filtered through a cellulosic membrane filter (filter diameter 47 mM, pore diameter 0.22 μ m in stainless steel filter holder). Each membrane filter was serially diluted up to 10^{-5} dilutions with 0.85% (w/v) NaCl solution and 100 μ l aliquots from each dilution were spread on Luria–Bertani agar plate, HiCrome™ *Klebsiella* Selective Agar (HiMedia) plate with and without meropenem (4 μ g ml^{-1}) and incubated at 37°C for 24 h. The dark purple colonies from meropenem-supplemented medium were selected and further passaged for 10 days on Luria–Bertani agar plates supplemented with meropenem to obtain stable carbapenemase producers. The isolates that exhibited stable resistance to carbapenems after 10 days were selected and further identified through VITEK 2 system at our teaching hospital and further confirmed by the PCR amplification of the *Khe* gene (Zhang *et al.* 2018). *Klebsiella pneumoniae* ATCC13883 was used as a control strain. ERIC-PCR fingerprinting was performed to determine the clonal relatedness of selected CRKP isolates according to Versalovic *et al.* (1991) to exclude the clonal strain. The identical ERIC-PCR profile of CRKP isolates originated from different sampling sites was included in further microbial analysis.

Antimicrobial susceptibility screening and multiple antibiotic resistance indexes

An antimicrobial susceptibility study was conducted by the Kirby–Bauer disk diffusion method. In this study, we used 17 antibiotics from 12 different antimicrobial classes: β -lactam–piperacillin (PI; 100 μ g); β -lactam/ β -lactamase inhibitor combinations—amoxicillin-clavulanic acid (AMC; 20/10 μ g) and piperacillin-tazobactam (PIT; 100/10 μ g); third-generation cephalosporin—ceftriaxone (CTR; 30 μ g) and ceftazidime (CAZ; 30 μ g); fourth-generation cephalosporin—cefepime (CPM; 50 μ g); carbapenems—meropenem (MRP; 10 μ g) and imipenem (IMP; 10 μ g); aminoglycosides—amikacin (AK; 10 μ g) and gentamicin (GEN; 10); tetracycline—tetracycline (TE; 30 μ g); fluoroquinolones—ciprofloxacin (CIP; 5 μ g) and levofloxacin (LE; 5 μ g); sulphonamides—co-trimoxazole (COT; 25 μ g); phenicol—chloramphenicol (C; 25 μ g); polymyxin—colistin (CL; 10 μ g); and glycycline—tigecycline (TGC; 15 μ g). In addition, the antimicrobial susceptibility study of colistin and tigecycline was performed by the broth dilution method. The results were interpreted as per the Clinical and Laboratory Standards Institute (CLSI 2020). The inhibition zone diameter and Minimum inhibitory concentrations (MIC) breakpoint of tigecycline were not mentioned in both CLSI and European Committee on Antimicrobial Susceptibility Testing (EUCAST); thus, the breakpoint of *Escherichia coli* was used for *K. pneumoniae*. If the isolate was resistant to

three or more antibiotic classes, it was classified as MDR and resistant to all classes but susceptible to one or two classes classified as XDR (Magiorakos *et al.* 2012). The percentages of resistant and susceptible isolates were determined as described by Smyth *et al.* (2020). The MAR index of each sampling site was determined using a formula based on the resistance profile (Krumperman 1983).

Molecular detection of carbapenemases, ESBL, and outer-membrane porin genes

A loopful of bacterial culture was suspended in 0.5 ml sterile water and heated at 95°C for 10 min before centrifuging at 10 000 × *g* for 15 min at 4°C to extract genomic DNA. The cell-free extract was used as template DNA and stored at −20°C for further analysis. The CRKP isolates were screened for carbapenemase (*bla*_{KPC}, *bla*_{NDM}, *bla*_{VIM}, *bla*_{IMP}, and *bla*_{OXA-48-like}), ESBL (*bla*_{TEM}, *bla*_{SHV}, and *bla*_{CTX-M}), and outer-membrane porin genes (*OmpK35*, *OmpK36*) through a PCR-based method using gene-specific primers (Table S6). In brief, 2 µl of the cell-free extract was combined with a PCR master mix containing 12.5 µl of 2× Taq PCR master mix (QIAGEN, India), 1 µl each of forward and reverse primers, and 8.5 µl of nuclease-free water. PCR amplifications were performed with Bio-Rad T100™ Thermal Cycler with the following thermocycling conditions: initial denaturation at 94°C for 3 min; a denaturation step of 30 cycles of denaturation at 94°C for 1 min; and annealing at 55°C for 45 s, an extension at 72°C for 1 min and a final elongation step at 72°C for 10 min. Annealing temperatures varied according to the melting temperatures of primers. The PCR products (5 µl) with 2 µl of the loading dye (Thermo Fisher Scientific) were separated electrophoretically at 50 V for 60 min in 1.2% agarose gel (Sigma) stained with ethidium bromide (10 µg/ml).

Molecular detection of virulence-associated genes

Six capsular serotypes encoding genes (K1 type, K2 type, K5 type, K20 type, K54 type, K57 type) and virulence-associated genes for siderophore production (*ybtS*, *irp-1*, *iroB*, *iucA*, *iutA*, *entB*), host adherence (*fimH*, *mrkD*, *kpcA*) and capsule upregulation (*rmpA* and *rmpA2*) were assessed by PCR. The nucleotide sequence of the primers and their melting temperatures are given in the supplementary table (Table S6).

Data analysis

Pearson correlation was used to determine the correlation between the physicochemical parameters, metals, and CFU count of CRKP of water samples. The correlation coefficient (*r*) was calculated, and plots were generated using R studio (package: corrplot) (Singh *et al.* 2019a).

Based on ERIC-PCR fingerprinting data, the presence and absence of DNA banding pattern were scored as 1 and 0, respectively. The scored binary data (1 and 0) were hierarchical agglomerative clustered to construct phylogenetic tree based on the Gower distance, which was calculated by daisy function of cluster R package (Gower 1971). The cluster analysis was performed by unweighted pair group method with arithmetic means using hclust function of stats R package. The silhouette coefficient was used to calculate the optimal number of clusters, and the similarity between them was calculated using cluster.stats function of fpc R package (<https://github.com/cran/fpc>). The dendrogram was visualized and inferred in FigTree v1.4.4 (Rambaut 2010).

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Supplementary data

Supplementary data is available at LAMBIO Journal online.

Conflict of interest

The authors declare no competing interest.

Author contributions

S.S.: writing—original draft preparation, data curation, investigation, methodology, experimentation work; R.K.S.: conceptualization, experimentation work, review and editing; M.G.: data curation; D.U.B.: data curation; A.S.: conceptualization; A.D.: writing, editing, experimentation work; S.U.D.: experimentation work; S.A.D.: data curation; E.S.: conceptualization, supervision, writing, review, editing, and resources.

Data Availability

The data that support the findings of this study are available in the supplementary material of this article.

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